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Application and Study of New Supramolecular Temporary Plugging Technology in Secondary Stimulation and Productivity Enhancement of Gas Wells

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Abstract

A supramolecular temporary plugging agent has been developed specifically for the low-porosity, low-permeability, fractured sandstone reservoirs in the Kuqa foreland of the Tarim Basin, where hydraulic fracturing is required to sustain stable long-term production of wells. Unlike conventional temporary plugging agents— which are unable to effectively plug fractures of 6 mm or larger and tend to cause reservoir contamination—this newly developed agent offers superior plugging performance without harming the reservoir. In addition, conventional temporary plugging agents can only achieve an inter-layer diversion pump pressure increase of 6–7 MPa, whereas the supramolecular temporary plugging agent achieves diversion pump pressures of 9.7–11.1 MPa.

Supramolecular materials are assembled through ternary combination relationships using the "bottom-up" approach of modern nanotechnology at the molecular level. Under ambient temperature conditions (20°C), the supramolecular temporary plugging agent exists as a low-viscosity liquid, facilitating injection. After being injected into the fracture, the agent undergoes a phase transition to form a solid-like supramolecular material as the temperature rises to a certain range (30–40°C for low-temperature systems; 80–100°C for high-temperature systems), This temporarily seals the fracture and diverts the fracturing fluid, leading to the formation of new fractures. With a further increase in temperature to 60–80°C (low-temperature systems) or 120–170°C (high-temperature systems), the material reverts to a liquid state and can be flowed back to the surface.

Based on the performance experiments of supramolecular temporary plugging agents, simulation of the operational temperature field, and results from field applications, this technology represents a significant advancement over conventional fracture steering methods, simplifying the implementation of volumetric fracturing or fracture network fracturing (including acid fracturing). Compared with traditional granular or fiber-based temporary plugging technologies, supramolecular temporary plugging agents require simpler pumping equipment and achieve superior fracture plugging effectiveness. A single hydraulic fracturing operation can generate multiple oil and gas flow channels within the fractures, thereby attaining the intended effects of volumetric or fracture network fracturing. After completion, the agent undergoes automatic

unblocking through thermally induced phase transitions, which causes no harm to the reservoir while significantly enhancing the productivity of individual wells.

This technology has been applied in over 10 gas wells in front of Kuqa Mountain in Tarim Oilfield. Compared with conventional temporary plugging agents, the plugging pressure is about 50% higher, which enables fracture creation further away from the wellbore after diversion and results in superior production performance post-treatment. This technological achievement holds broad prospects for application and popularization, providing valuable reference for gas fields in the Kuqa Mountain foreland and other tight sandstone reservoirs.

Introduction

The Keshen Gas Field, located in the Kuqa foreland of the Tarim Basin, has reservoirs with an average burial depth exceeding 6,500 meters and formation temperatures ranging from 145°C to 190°C. The measured core porosity of these reservoirs mainly falls within the range of 5% to 7%, with an average porosity of 5.52%. The measured matrix permeability is primarily between (0.1–0.5) × 10^{-3} μm^2, averaging 0.56 × 10^{-3} μm^2, characterizing the reservoirs as low-porosity, low-permeability fractured sandstones[1]. Due to poor and heterogeneous reservoir properties and significant variations in vertical stress, particle-size temporary plugging and diverting proppant fracturing has become one of the primary measures to enhance the stimulation of these ultra-thick reservoirs in the Kuqa foreland area[2].

In recent years, as the development of several gas reservoirs in the Keshen Gas Field has entered the mid-to-late stages, more severe challenges have arisen. Firstly, for some wells, production declines rapidly after water breakthrough, even resulting in a complete loss of productivity. Geological studies indicate that these wells generally contain substantial volumes of untapped residual gas[3]. To address the need to further exploit residual gas in old wells, new fractures are created through diversion fracturing techniques. Secondly, water-producing wells and drainage wells generally display high-mineralization water, accelerating scaling in both the formation and wellbore[4–5]. In such cases, conventional wellbore descaling methods have only short-term effectiveness, and fracturing modifications are therefore required to maintain stable and long-term single-well production.

Field practice has shown that water produced from gas wells in the Keshen Block commonly contains formation sands or silty fill, indicating reservoir depletion. Current temporary plugging agents Figure 1[4] have the following limitations: fiber and solid particle temporary plugging techniques impose higher requirements on surface pumping equipment and personnel, and block only specific "point" or "line" locations. While single fibers can temporarily plug fractures with widths of 2 mm or 4 mm, they are ineffective for fractures 6 mm wide or larger[5]. Statistical analysis of numerous fracturing curves demonstrates that fiber and solid particle temporary plugging agents can only increase interlayer diversion pump pressure by 6–7 MPa, and these agents are often non-degradable or difficult to remove. Additionally, gel-based and natural plant-based plugging agents can inflict secondary damage to the reservoir and are difficult to break down[5]. These issues complicate and challenge the selection of secondary fracturing technologies for old wells. Therefore, there is an urgent need to develop low-damage temporary plugging technologies capable of effectively sealing previously created fractures to form new fractures, or generating higher static pressures within existing fractures to induce the creation of new fracture systems.

Figure 1—Commonly used diverting materials in chemical diversion technology

Supramolecular temporary plugging diverting agent development

Theoretical basis

Supramolecular chemistry, defined as "the chemistry of molecular assemblies and intermolecular bonds," has, through its integration with materials science, life sciences, and other disciplines, developed into supramolecular science—considered an important source of emerging concepts and high technologies in the 21st century. Supramolecular materials not only possess many properties of conventional materials, but also exhibit special properties that are difficult to achieve with traditional materials. The polymerization and degradation of supramolecular materials can reversibly occur, endowing them with properties such as low-temperature processability, self-healing, and responsiveness to environmental stimuli[5–6]. Thus, they are regarded as "smart materials."

Supramolecular temporary plugging diverting agent construction

Based on supramolecular self-assembly theory, a liquid temporary plugging diverting agent with a unique phase-change process was synthesized using lithium chloride, β-cyclodextrin, $C_8H_{17}OH$ (octanol), methyl cellulose, and N-methylformamide at a mass ratio of 0.5:15:4:0.2:80.3. As shown in Figure 2, the agent is liquid at ambient temperature, and undergoes a phase transition (with tunable phase transition temperature) as temperature changes. When the agent is injected into the fracture, it transitions from liquid to solid as the temperature rises to the solidification point, temporarily plugging the fracture and diverting the fracturing fluid to open new fractures[5–6]. As the temperature within the fracture further increases, the solid supramolecular material reverts to a liquid state, allowing for flowback to the surface following the operation.

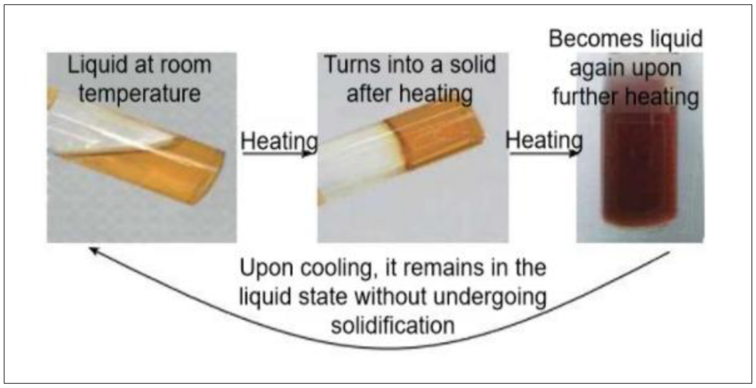

Figure 2—Phase transition process of liquid temporary plugging diverting agent

Compared with traditional particle or fiber-based temporary plugging methods, the supramolecular temporary plugging agent requires simpler pumping equipment and provides superior fracture plugging performance. A single hydraulic fracturing stage can generate multiple oil and gas permeable fracture pathways, realizing the effect of volumetric or fracture network fracturing. After the operation, the plug is removed automatically via temperature-controlled phase transition, causing no damage to the reservoir and thus achieving a significant increase in single-well productivity.

Performance evaluation of supramolecular temporary plugging diverting agent

Gel formation and gel breaking properties

High-temperature and low-temperature systems of supramolecular temporary plugging diverting agent have been developed according to different reservoir temperatures. The temperature–phase behavior of the liquid diverting agent systems was studied to determine the temperature intervals corresponding to different phase transitions. Liquid diverting agents were divided into high-temperature and low-temperature categories and tested separately.

High-temperature system. Phase transition experiment: The liquid diverting agent at room temperature was heated. When the temperature reached 60°C, a phase transition began, and complete solidification required approximately 17 to 25 minutes. In the temperature range of 80–100°C, the solid-phase transition of the liquid diverting agent occurred within 5 minutes, with complete solidification reached within 12 minutes. In the 110–120°C interval, phase transition initiation and solidification still occurred, with a total process time of about 8 minutes. When the temperature exceeded 120°C, the solid diverting agent reverted to a liquid state (Figure 3).

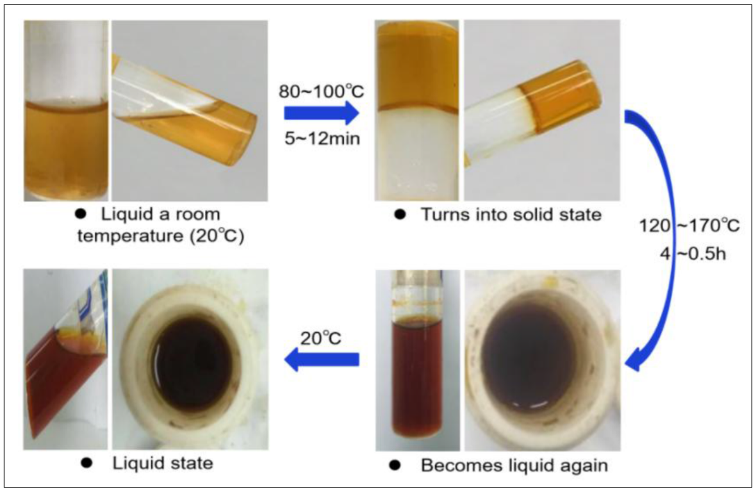

Figure 3—Phase transition experiment of High-temperature systems

Effect of temperature on phase transition: Experimental results showed (Figure 4) that as temperature increased, the onset and duration of phase transition and solidification of the liquid diverting agent shortened. The phase transition temperature interval was determined as 80–120°C, while the liquefaction temperature interval was 120–170°C. With increasing temperature, the volume of the liquid diverting agent increased, and solidification time slightly increased (Figure 5). The high-temperature system is suitable for reservoirs with temperatures above 120°C.

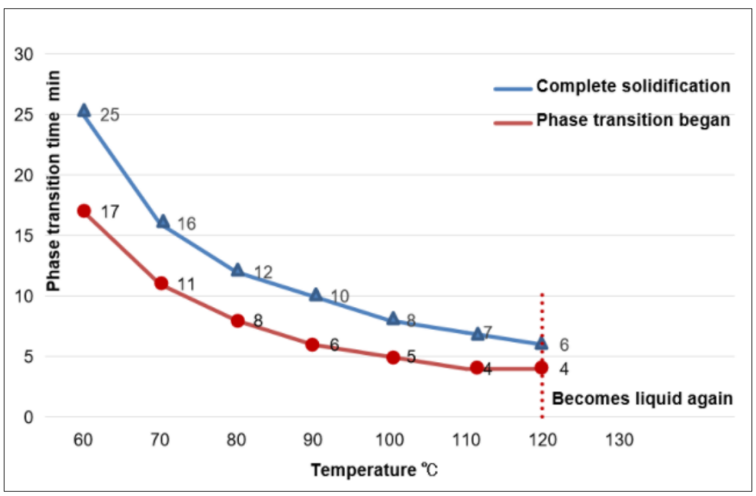

Figure 4—Relationship between the phase transition states of liquid diverting agents and temperature (high-temperature systems)

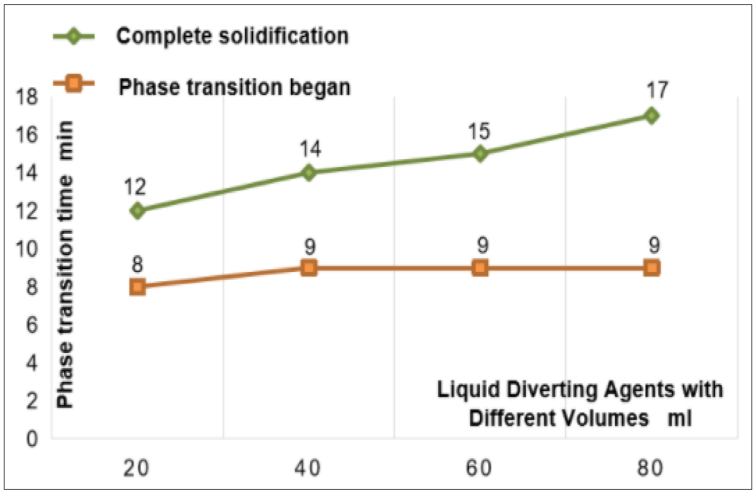

Figure 5—Phase transition time of liquid diverting agents with different volumes at 80#

Low-Temperature System. Phase transition experiment: The low-temperature system liquid diverting agent was heated from room temperature. When the temperature reached 30°C, the phase transition to a solid occurred in 11 minutes, with complete solidification in 22 minutes. At 40°C, the starting point of the phase transition was at 7.5 minutes, with complete solidification in 15 minutes. When the temperature reached 60–80°C, the diverting agent began to liquefy, At 80°C, the liquefaction degree reaches 40% after 2 hours, and complete liquefaction is achieved after 180 hours (Figure 6, Figure 7).

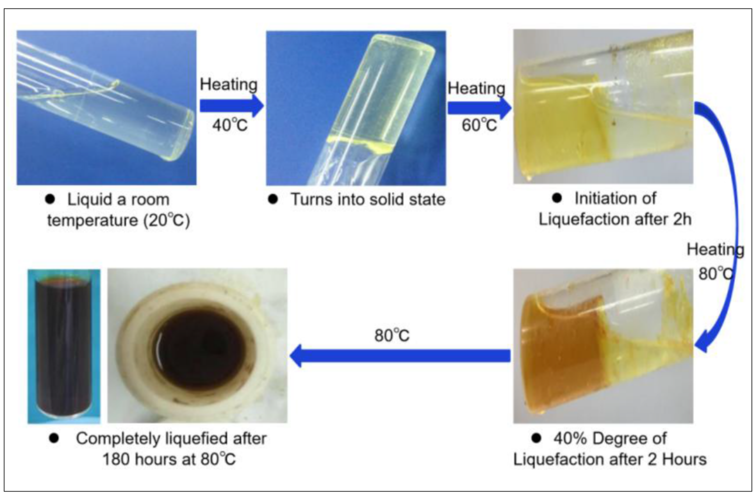

Figure 6—Phase transition experiment of Low-temperature systems

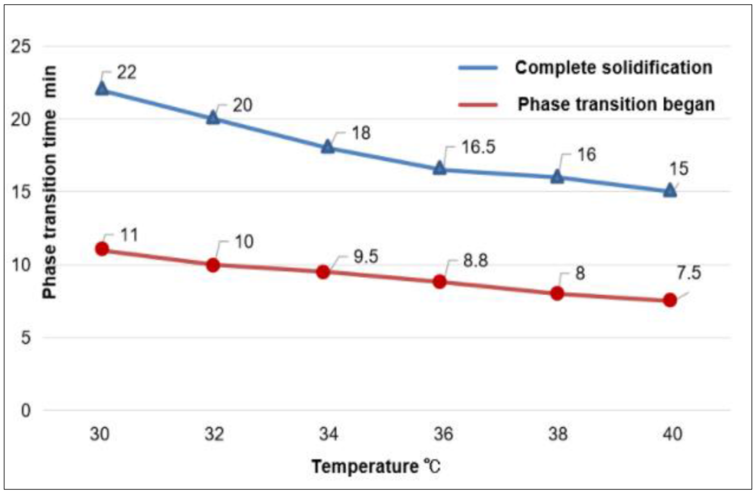

Figure 7—Relationship between the phase transition states of liquid diverting agents and temperature (low-temperature systems)

Effect of temperature on phase transition: The experimental results showed that as the temperature increased, the time required for initiation and completion of phase transition and solidification decreased. The phase transition interval was 30–60°C, while the liquefaction interval was 60–80°C. The volume of the liquid diverting agent increased with temperature, and the solidification time slightly increased (Figure 8). The low-temperature system is suitable for reservoirs with temperatures below 120°C.

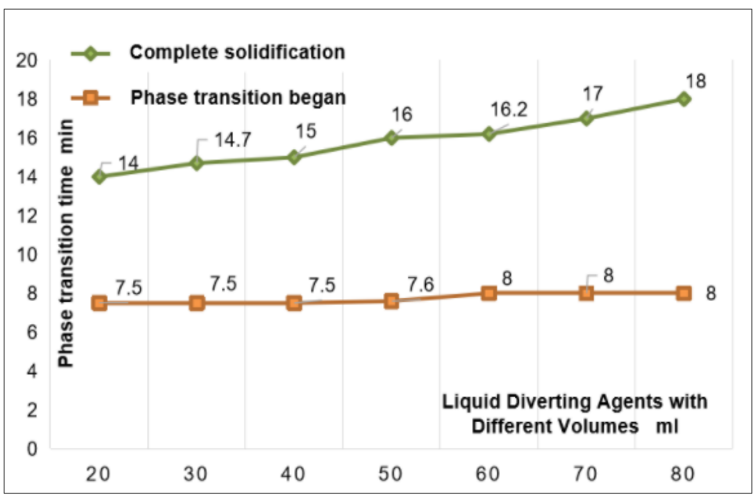

Figure 8—Phase transition time of liquid diverting agents with different volumes at 40#

Based on the above experiments, it can be concluded that temperature is a critical factor controlling the phase transition and solidification of liquid diverting agents. The experimental results indicate that the temporary plugging agent can autonomously decompose into a liquid phase when triggered by temperature. The resultant liquid is free of residues, thus minimizing reservoir damage. By testing the time required for solidification and correlating it with the wellbore temperature distribution, injection parameters can be designed to ensure that solidification does not occur within the wellbore.

Plugging and Diverting Performance

Displacement pressure experiments were conducted using a constant-flow pump to simulate pipe flow devices. The displacement flow rate was set at 3.8 ml/min, determined by calculating the linear velocity within the fracture based on typical hydraulic fracturing rates. Three different pipe lengths—30 cm, 75 cm, and 150 cm—were selected, with a pipe radius of 5 mm. The experiments were carried out at two temperatures: 90°C (high-temperature system) and 60°C (low-temperature system).

In the high-temperature system, experimental results from three groups revealed the time required for the liquid diverting agent to fully solidify and the corresponding maximum pressure resistance for each pipe length. For the 30 cm pipe, complete solidification occurred in 5 minutes, with a maximum pressure resistance of 3.1 MPa; for the 75 cm pipe, complete solidification took 5.5 minutes, with a maximum pressure resistance of 6.6 MPa; and for the 150 cm pipe, complete solidification was achieved in 6 minutes, with a maximum pressure resistance of 15.5 MPa (Figures 9–11). These results were used to construct a relationship curve between pipe length and pressure resistance (Figure 12).

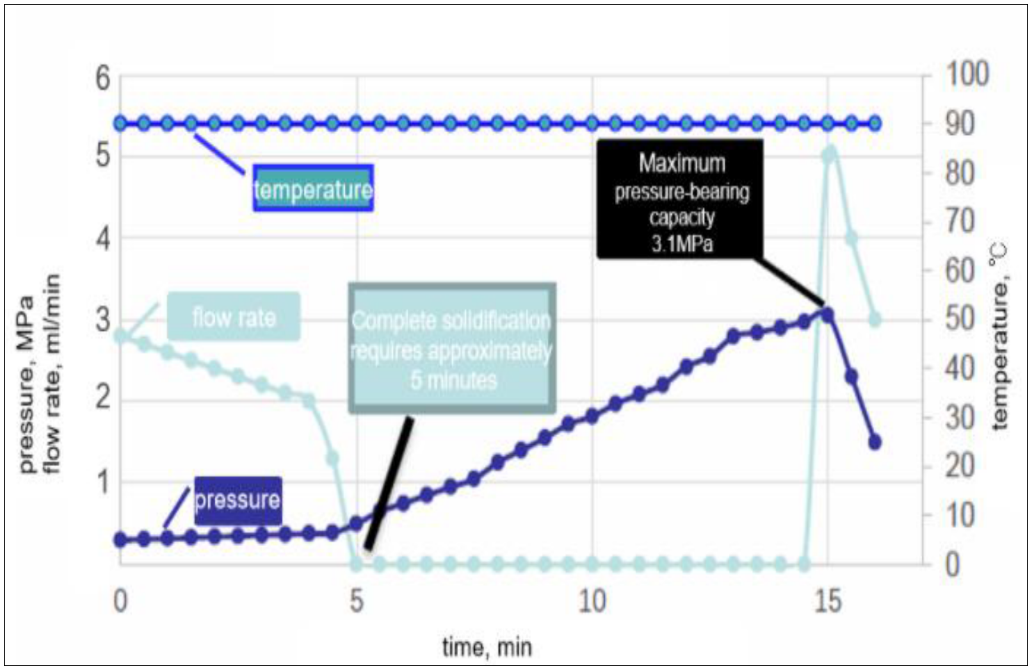

Figure 9—Pressure resistance curve for 30cm pipe at 90°C

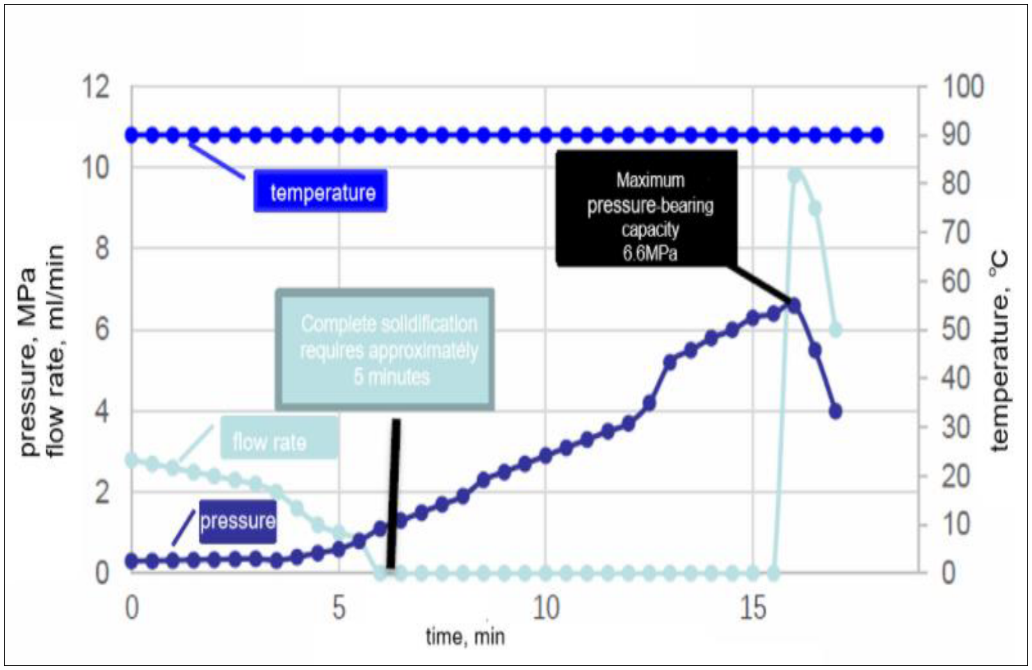

Figure 10—Pressure resistance curve for 75cm pipe at 90°C

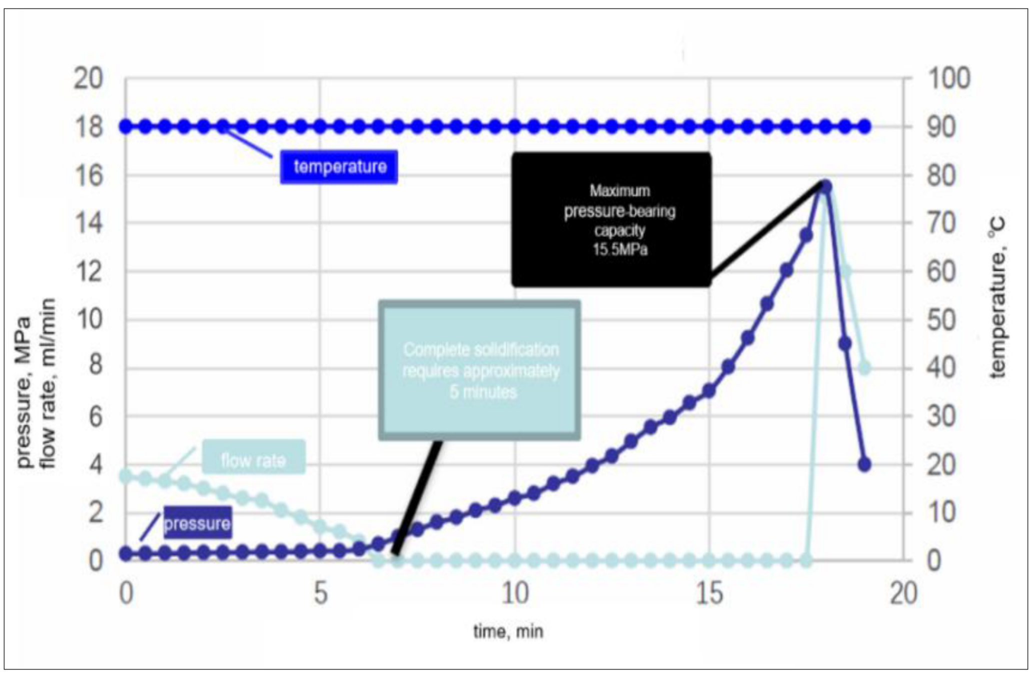

Figure 11—Pressure resistance curve for 150cm pipe at 90°C

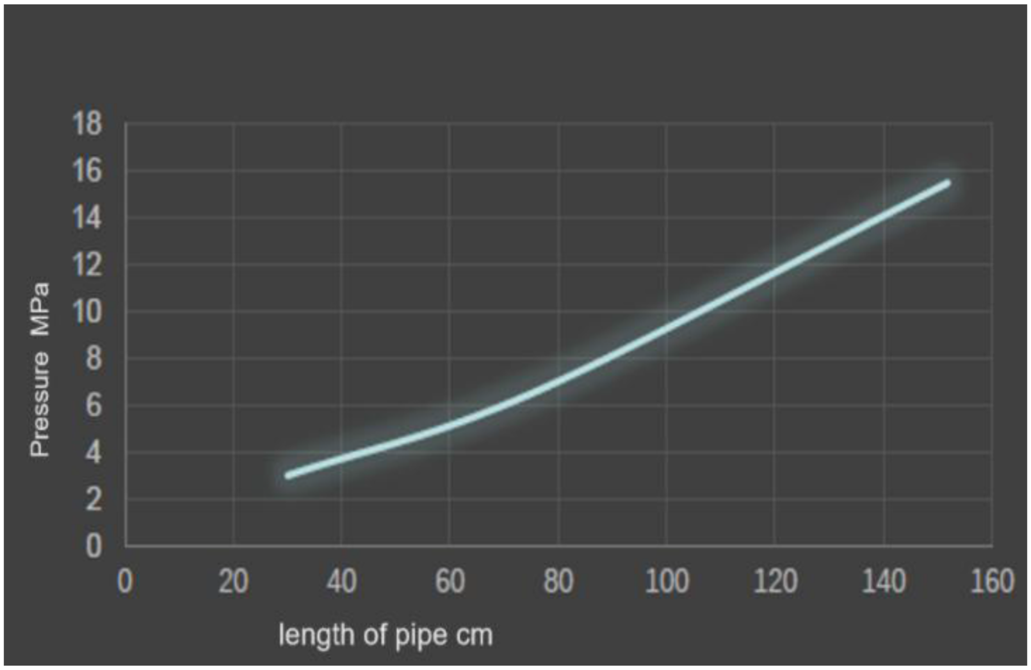

Figure 12—Relationship curve between pipe length and pressure resistance (high-temperature system)

Similarly, in the low-temperature system, the required times for the liquid diverting agent to attain a fully solidified state and the associated maximum pressure resistances for the different pipe lengths were 5 minutes/2.2 MPa, 5.5 minutes/4.9 MPa, and 6 minutes/11 MPa, respectively (Figures 13–15). Based on these findings, the relationship between pipe length and pressure resistance was also plotted (Figure 16).

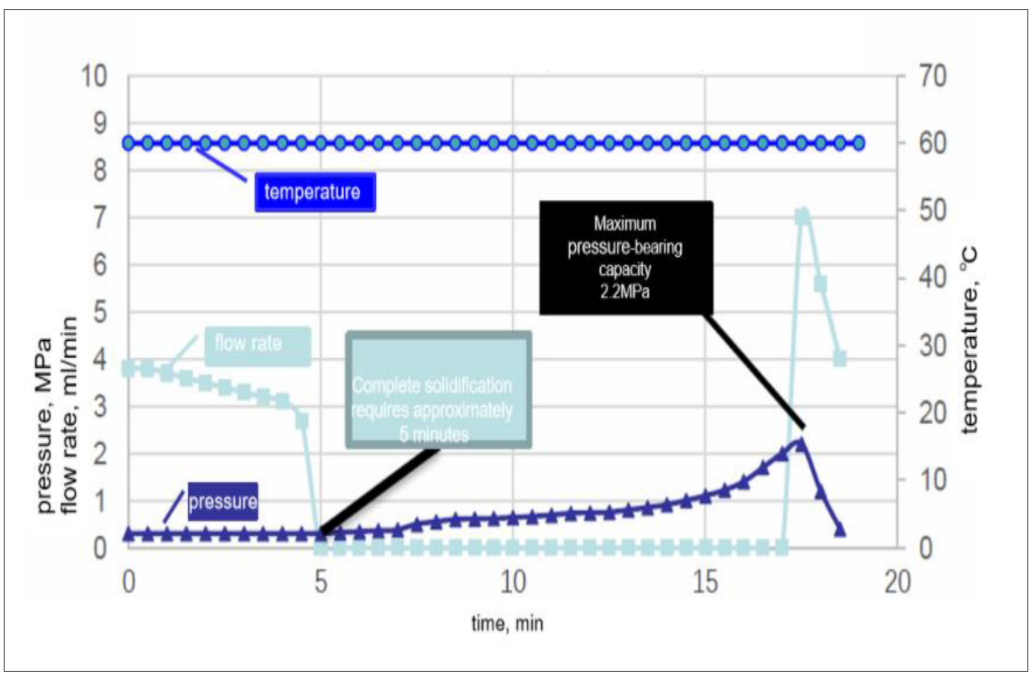

Figure 13—Pressure resistance curve for 30cm pipe at 60°C

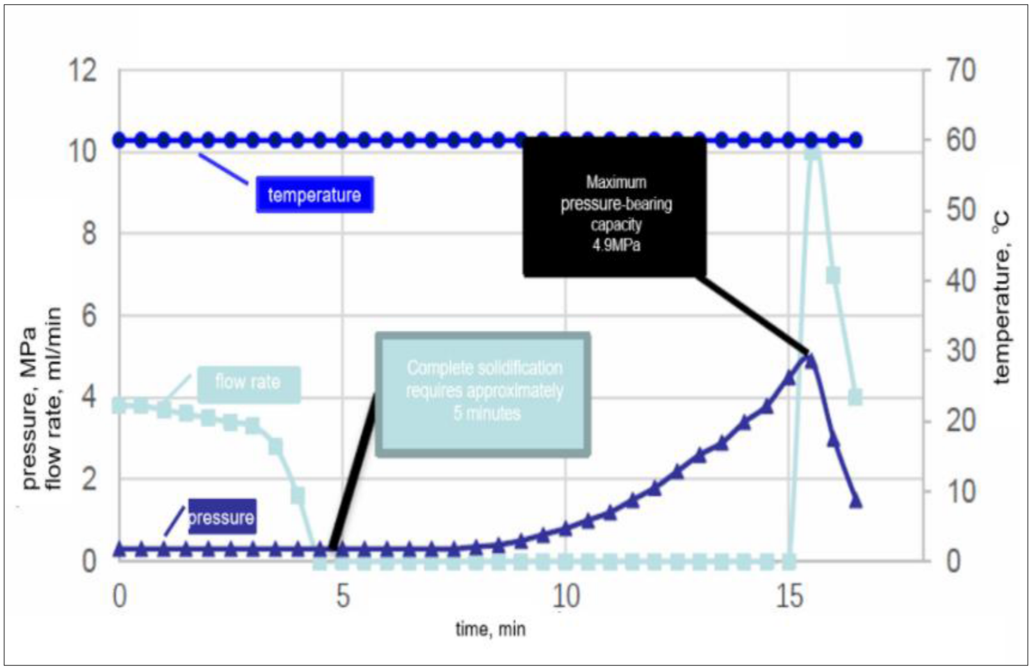

Figure 14—Pressure resistance curve for 75cm pipe at 60°C

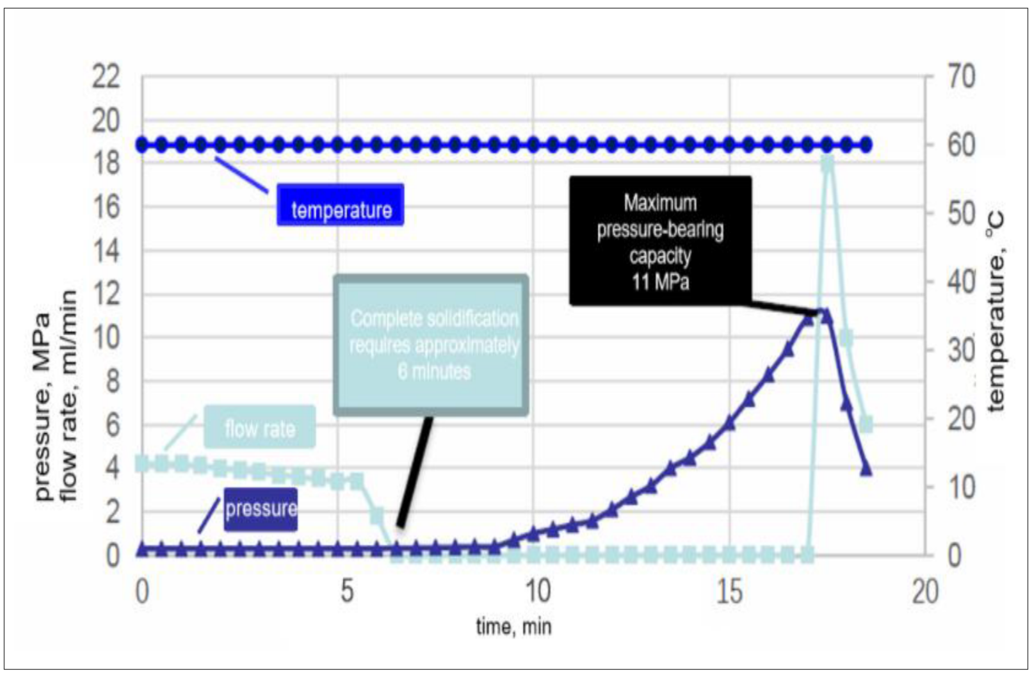

Figure 15—Pressure resistance curve for 150cm pipe at 60°C

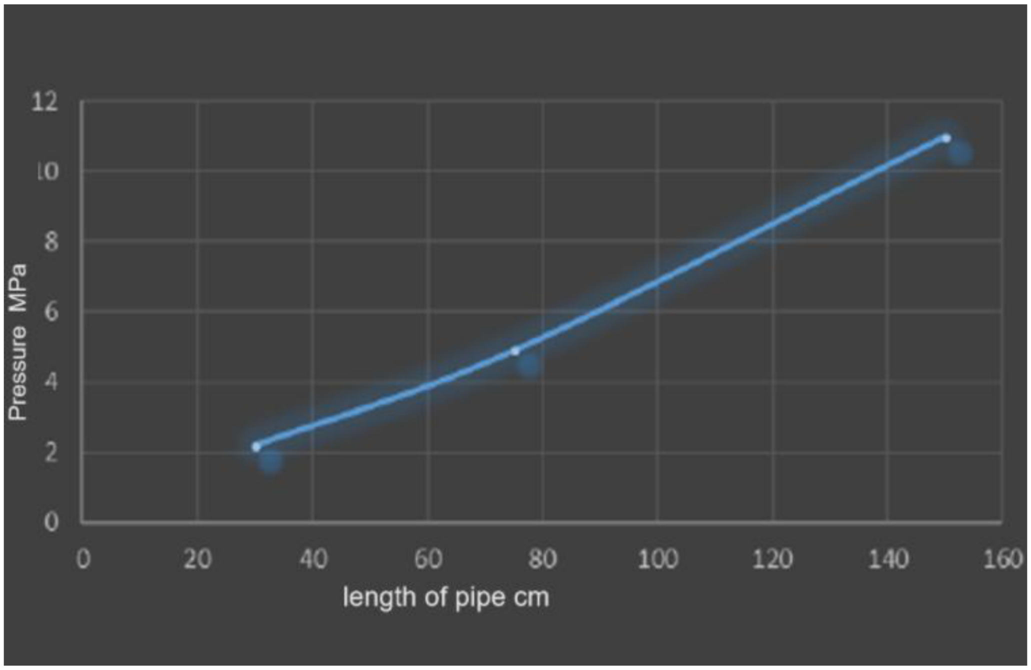

Figure 16—Relationship curve between pipe length and pressure resistance (low-temperature system)

Wellbore Temperature Field Simulation Evaluation

A wellbore temperature field simulation was conducted for a single well in the Keshen II block of the Kuqa Foreland to calculate the temperature distribution in the wellbore at displacement rates of 6.0 m^3/min. Prior to the injection of the supramolecular diverting agent, the total design fluid volume was 600 m^3. The reservoir temperature in the stimulated interval was 152.55 °C. Simulation results showed that, after 600 m^3 of fracturing fluid entered the reservoir, both the wellbore and reservoir temperatures decreased, with the bottomhole temperature stabilizing at approximately 58°C (Figure 17), meeting the operating conditions required for the supramolecular diverting agent.

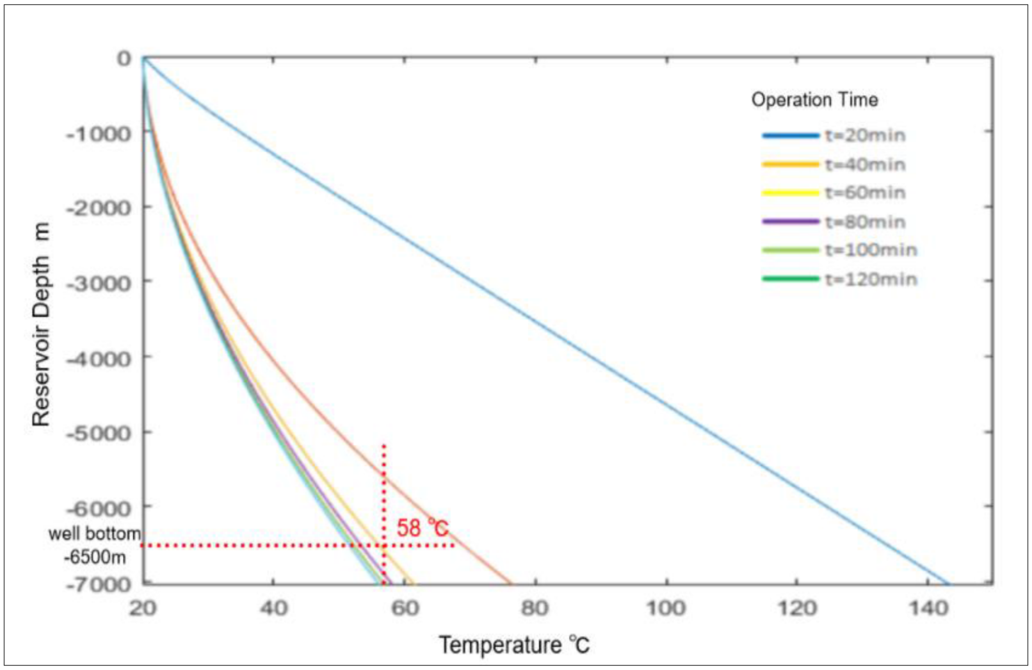

Figure 17—Wellbore temperature distribution at a displacement rate of 6 m^3/min under different injection durations

The supramolecular diverting agent remains liquid under ambient conditions. As temperature increases within the range of 60°C to 80°C, the complete solidification time of the agent varies from 5 to 25 minutes—the higher the temperature, the shorter the gelation time. To prevent premature solidification of the diverting agent within the wellbore, the allowable residence time of the agent in the wellbore is limited to 15 minutes. Therefore, after the injection of the supramolecular diverting agent is completed, the displacement rate should be increased immediately to minimize its residence time in the wellbore.

Field Application

A novel supramolecular temporary plugging agent achieved diversion pump pressures of 9.7–11.1 MPa during secondary fracturing operations in old wells in the Kuqa foreland. Compared with conventional temporary plugging agents, it exhibited superior blocking performance, with diversion pump pressures approximately 50% higher. Furthermore, it caused zero damage to the reservoir, demonstrating better overall performance and temporary plugging effectiveness.

Case Study 1

Well KS-801 was brought into production on August 11, 2015. On November 8, 2021, reservoir pressure dropped to zero and natural flow ceased; subsequently, continuous gas lift and coiled tubing drainage operations were initiated, with a flowing pressure of 1 MPa and an average daily drainage rate of 200 m^3/d. As a key drainage well in the western part of the Keshen 8 gas reservoir, KS-801 exhibited a cyclic decline in water production during drainage. Frequent deplugging operations were required to maintain drainage rates, with the shortest cycle between interventions being three months, resulting in high operational frequency and cost. Geological analysis indicated that this well operates as a "gas-water separate path" production well[1–2], with the lower section as the main water-producing interval and a substantial volume of remaining gas in the upper interval.

A supramolecular temporary plugging diverting and proppant fracturing operation was carried out in Well KS-801. The diversion pumping pressure increased by 11.1 MPa, indicating a significant temporary plugging effect. Following fracturing, the wellhead pressure reached 10.51 MPa, daily gas production increased by 4×10^4 m^3/d, and daily drainage reached 511 m^3(Figure 18), greatly enhancing single-well drainage performance and expanding the water control radius. This resulted in substantial economic benefits, such as stable single-well drainage and further recovery of remaining gas, far exceeding initial expectations. Additionally, the increased well drainage effectively delayed water breakthrough in mid-to-upstructure wells, laying a stronger foundation for stable production and supply from mature blocks.

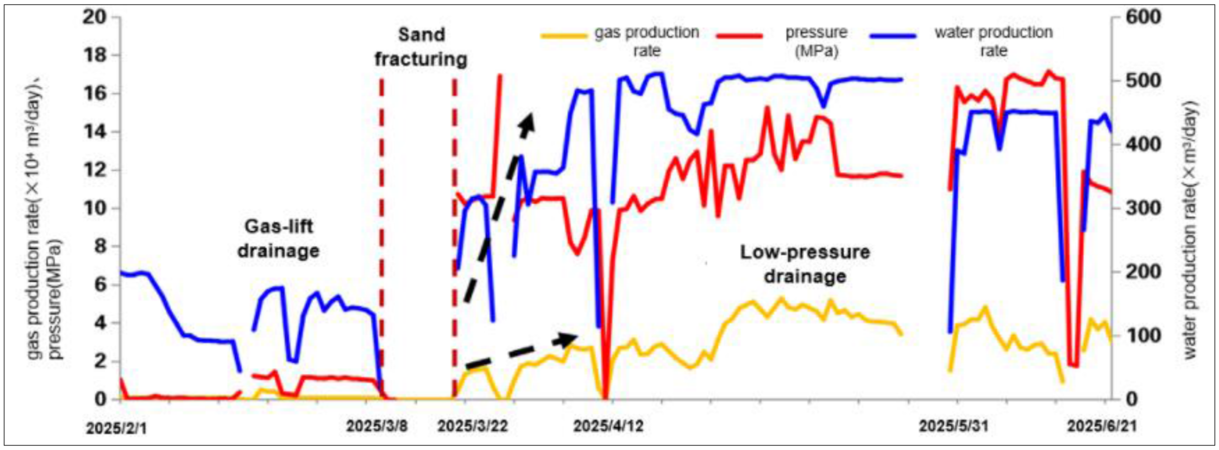

Figure 18—Production curves before and after supramolecular temporary plugging diverted sand fracturing (KS-801)

Case Study 2

Well KL-241-2 was brought into production on April 5, 2020. Water breakthrough was observed in August 2021, and from April 14, 2022, there was a sharp decline in production. The stimulated interval of this well spans 107.5 meters, with a minimum horizontal principal stress of 2.09 MPa/100m, a vertical principal stress of 2.45 MPa/100m, and a maximum horizontal principal stress ranging from 2.60 to 2.79 MPa/100m. The stress regime is SH > SV > Sh, characteristic of a strike-slip faulting system, which is favorable for the formation of fracture networks[1]. However, the large horizontal stress difference necessitates the use of temporary plugging diversion combined with proppant fracturing technology to enhance the vertical utilization of the reservoir, create high-conductivity artificial fractures, enlarge the effective drainage area, increase water-bearing production capacity, and delay water invasion in the mid-to-upstructure portions of the gas reservoir.

Supramolecular temporary plugging diverted proppant fracturing was performed in Well KL-241-2, resulting in a 9.7 MPa increase in diversion pump pressure, indicating a significant temporary plugging effect. After fracturing, the wellhead pressure increased from 31.1 to 33.6 MPa, daily drainage rose from 86 to 123 m^3/d, and daily gas production increased from 4.2 to 5.5 $\times 10^4$ m^3/d (Figure 19). The well has since maintained stable production after fracturing, with cumulative incremental gas production exceeding 10 million cubic meters.

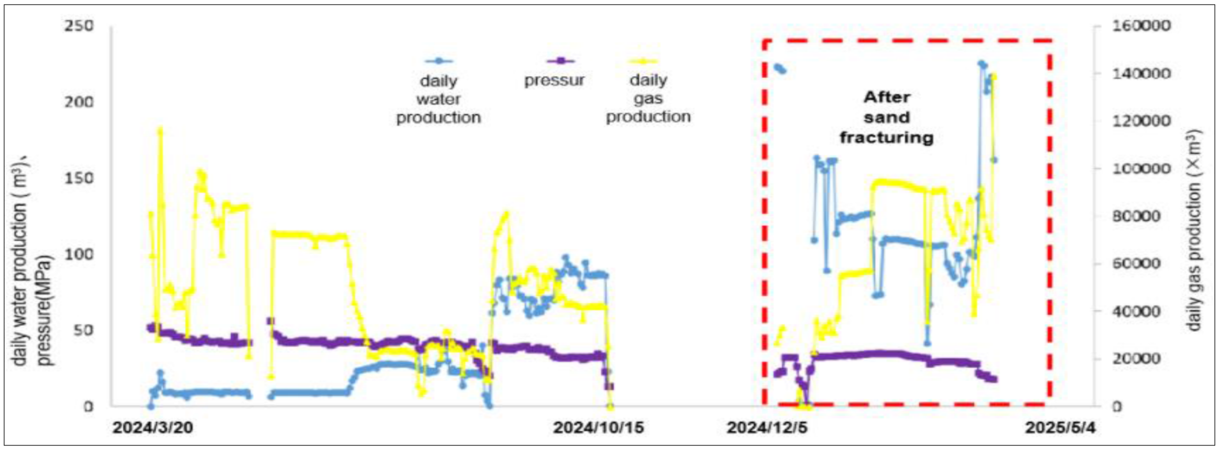

Figure 19—Production curves before and after supramolecular temporary plugging diverted sand fracturing (KL-241-2)

Conclusions

1. Based on supramolecular mechanisms, a novel supramolecular temporary plugging diversion agent has been developed. The liquid diversion agent system exhibits low viscosity at room temperature, providing good injectivity. This type of liquid can conveniently and completely seal fractures, capable of temporarily plugging fractures of 6 mm or wider compared to conventional temporary plugging agents. The plugging and unplugging times are precisely controllable and can be optimized by design. The system is a solid-free liquid system that undergoes automatic phase transition to a solid state under certain temperature conditions and reverts to a liquid upon further heating, without the need for additional additives or external diversion materials. It satisfies the technical requirements for diversion fracturing (diversion acid fracturing) and is an intelligent temporary plugging agent.
2. Practical application has demonstrated that during secondary fracturing operations in old wells in the Kuqa foreland, conventional temporary plugging agents can only achieve an inter-layer diversion pump pressure increase of 6–7 MPa, whereas the supramolecular temporary plugging agent achieves diversion pump pressures of 9.7–11.1 MPa. This technology breaks through traditional fracture diversion concepts, making volumetric fracturing or fracture network fracturing (acid fracturing) simpler and more feasible. Compared with conventional temporary plugging agents, it has superior plugging performance and is an ideal diversion agent for secondary fracturing and enhanced recovery of old wells in the Kuqa foreland tight sandstone gas reservoirs, provides valuable experience for similar reservoir fracturing operations, and has broad prospects for wider application.

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